

Qualification Pack



Advanced CNC Programmer

QP Code: TNSD/CSC/Q0501

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

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TNSD/CSC/Q0501: Advanced CNC Programmer

Brief Job Description

An Advanced CNC Programmer designs, develops, and optimizes complex CNC programs for various tool room machines, ensuring the precise manufacture of parts according to advanced specifications. This role involves in-depth knowledge of machining processes, CAD/CAM software, quality control, and production planning. Upon successful completion, learners will be qualified for advanced programming roles in engineering, manufacturing, machine tools, and automobile industries, enabling them to lead complex machining projects and contribute to strategic production goals.

Personal Attributes

Upon successful completion, learners will be qualified for advanced programming roles in engineering, manufacturing, machine tools, and automobile industries, enabling them to lead complex machining projects and contribute to strategic production goals.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TNSD/CSC/N6901: Perform Advanced Workshop Operations](#)
2. [TNSD/CSC/N6902: Implement Advanced CNC Programming Techniques](#)
3. [TNSD/CSC/N6903: Generate Complex CNC Toolpaths using Master cam](#)
4. [TNSD/CSC/N6904: Design Advanced 3D Models using SolidWorks](#)
5. [TNSD/CSC/N6905: Design and Fabricate Jigs and Fixtures](#)
6. [TNSD/CSC/N6906: Execute Quality Control and CMM Inspections](#)
7. [TNSD/CSC/N6907: Apply Principles of Strength of Materials in Design](#)
8. [TNSD/CSC/N6908: Estimate and Cost Manufacturing Projects](#)
9. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)
10. [TNSD/CSC/N6909: Implement Production Planning and Control Strategies](#)
11. [TNSD/CSC/N6910: Apply Principles of Management in a Manufacturing Environment](#)
12. [TNSD/CSC/N0501: OJT\(INDUSTRY\)](#)

Qualification Pack

Qualification Pack (QP) Parameters

Sector	Capital Goods
Sub-Sector	
Occupation	CNC Programmer
Country	India
NSQF Level	4.5
Credits	40
Aligned to NCO/ISCO/ISIC Code	7233.0200
Minimum Educational Qualification & Experience	Previous relevant Qualification of NSQF Level (4) OR Previous relevant Qualification of NSQF Level (Precision Machinist-CNC(Level 4))
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-4.5-CG-04456-2025-V1-TNSDC
NQR Version	1.0

Qualification Pack

TNSD/CSC/N6901: Perform Advanced Workshop Operations

Description

Understand the programming and manufacturing of complex parts by using CNC Turn Mill and CAM software. Understand the programming and manufacturing of complex parts by using 5-Axis CNC Milling Machine and CAM software.

Scope

The scope covers the following :

- Understand the manufacturing of complex parts by using CNC Turn mill, 5-Axis CNC Milling.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate the safety aspects and Special modules on CNC Turn Mill and CNC 5-Axis Milling.
- PC2.** Practice on Different parts and its functions of CNC Turn Mill and CNC 5-Axis Milling.
- PC3.** Practice on Exercises with proper cutting terminology on CNC Turn Mill.
- PC4.** Practice on Exercises with proper cutting terminology on CNC 5-Axis Milling.
- PC5.** Demonstrate Part programming and Machining of parts on CNC Turn mill.
- PC6.** Demonstrate Part programming and Machining of parts on CNC 5-Axis Milling
- PC7.** Machine different parts of a project on CNC machines and assembling/OJT
- PC8.** Practice measuring training on CMM

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Practice on Different parts and its functions of CNC Turn Mill and CNC 5-Axis Milling.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Part programming and Machining of parts on CNC Turn mill.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	100	75	25
PC1. Demonstrate the safety aspects and Special modules on CNC Turn Mill and CNC 5-Axis Milling.	-	10	-	-
PC2. Practice on Different parts and its functions of CNC Turn Mill and CNC 5-Axis Milling.	-	10	-	-
PC3. Practice on Exercises with proper cutting terminology on CNC Turn Mill.	-	15	-	-
PC4. Practice on Exercises with proper cutting terminology on CNC 5-Axis Milling.	-	15	-	-
PC5. Demonstrate Part programming and Machining of parts on CNC Turn mill.	-	20	-	-
PC6. Demonstrate Part programming and Machining of parts on CNC 5-Axis Milling	-	20	-	-
PC7. Machine different parts of a project on CNC machines and assembling/OJT	-	-	75	25
PC8. Practice measuring training on CMM	-	10	-	-
NOS Total	-	100	75	25

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Not
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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6901
NOS Name	Perform Advanced Workshop Operations
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, Capital Goods
NSQF Level	4.5
Credits	12
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N6902: Implement Advanced CNC Programming Techniques

Description

Able to do the programming and manufacturing of components on Multi-Axis CNC lathe. Able to do the programming and manufacturing of components on 5-Axis Milling machine.

Scope

The scope covers the following :

- Understand the concepts of Multi-Axial CNC machines.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the Basic Concepts and constructional features of CNC Turn mill and CNC 5-Axis milling.
- PC2.** understand the Axis movements and limitation
- PC3.** understand the Concept of tool management, work offset & coordinates system
- PC4.** Learn the Part programming
- PC5.** Understand the DNC setup ,program copy and execution
- PC6.** Program and perform machining on CNC Turn mill.
- PC7.** Program and do machining on CNC 5-Axis Milling.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** understand the Axis movements and limitation

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Program and perform machining on CNC Turn mill.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	90	-	10
PC1. Learn the Basic Concepts and constructional features of CNC Turn mill and CNC 5-Axis milling.	10	-	-	-
PC2. understand the Axis movements and limitation	10	-	-	-
PC3. understand the Concept of tool management, work offset & coordinates system	30	-	-	-
PC4. Learn the Part programming	20	-	-	-
PC5. Understand the DNC setup ,program copy and execution	30	-	-	-
PC6. Program and perform machining on CNC Turn mill.	-	45	-	5
PC7. Program and do machining on CNC 5-Axis Milling.	-	45	-	5
NOS Total	100	90	-	10

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6902
NOS Name	Implement Advanced CNC Programming Techniques
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, CNC Programmer
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N6903: Generate Complex CNC Toolpaths using Master cam

Description

Simulation of CNC program by Master cam software before machining and doing the actual machining of a simple irregular profile.

Scope

The scope covers the following :

- Understand the generation of NC program and transferring it to machine.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate the Wireframe Creation
- PC2.** Demonstrate Editing and Analyzing geometry
- PC3.** Demonstrate Transforming geometry
- PC4.** Do 3D Solid creation
- PC5.** Demonstrate tool path identification

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Demonstrate Editing and Analyzing geometry
- KU2.** Demonstrate tool path identification

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Transforming geometry

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	85	-	15
PC1. Demonstrate the Wireframe Creation	-	15	-	5
PC2. Demonstrate Editing and Analyzing geometry	-	15	-	5
PC3. Demonstrate Transforming geometry	-	20	-	-
PC4. Do 3D Solid creation	-	15	-	5
PC5. Demonstrate tool path identification	-	20	-	-
NOS Total	-	85	-	15

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6903
NOS Name	Generate Complex CNC Toolpaths using Master cam
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, CNC Programmer
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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TNSD/CSC/N6904: Design Advanced 3D Models using SolidWorks

Description

Modelling of Machine parts/ Jigs and Fixtures using solid works. Assembling of Machine parts/ Jigs and Fixtures using solid works.

Scope

The scope covers the following :

- Understand the different modules used in assembly drawing and sheet metal processing.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate different tools used in solid works.

PC2. Demonstrate different modules available in Solid works.

PC3. Create Models by using different tools and modules

PC4. Demonstrate Part assembly by using Solid works

PC5. Demonstrate drawing of different types of Jigs.

PC6. Demonstrate drawing of different types of Fixtures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Demonstrate Part assembly by using Solid works.

KU2. Demonstrate drawing of different types of Fixtures

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Demonstrate different tools used in solid works.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	100	-	-
PC1. Demonstrate different tools used in solid works.	-	15	-	-
PC2. Demonstrate different modules available in Solid works.	-	15	-	-
PC3. Create Models by using different tools and modules	-	10	-	-
PC4. Demonstrate Part assembly by using Solid works	-	10	-	-
PC5. Demonstrate drawing of different types of Jigs.	-	25	-	-
PC6. Demonstrate drawing of different types of Fixtures.	-	25	-	-
NOS Total	-	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6904
NOS Name	Design Advanced 3D Models using SolidWorks
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, CNC Programmer
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N6905: Design and Fabricate Jigs and Fixtures

Description

Able to understand the Functioning of Jig and fixture for mass production. Able to understand concept and drawing of Jigs & Fixtures.

Scope

The scope covers the following :

- Understand the different types of jigs and its parts used for production purpose.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** • Learn the Degrees of freedom, location of job and planes of movements in a
• jig, fixture.
- PC2.** Understand the Difference between jig and fixture and its parts.
- PC3.** Understand the concept of Tool location with respect to machine in a jig and fixture
- PC4.** Understand the Locating work piece in a Jig & Fixture.
- PC5.** Understand Mounting and location of a fixture on a machine tool

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Learn the Degrees of freedom, location of job and planes of movements in a
• jig, fixture.
- KU2.** Understand the concept of Tool location with respect to machine in a jig and fixture

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand Mounting and location of a fixture on a machine tool

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. • Learn the Degrees of freedom, location of job and planes of movements in a • jig, fixture.	20	-	-	-
PC2. Understand the Difference between jig and fixture and its parts.	20	-	-	-
PC3. Understand the concept of Tool location with respect to machine in a jig and fixture	20	-	-	-
PC4. Understand the Locating work piece in a Jig & Fixture.	20	-	-	-
PC5. Understand Mounting and location of a fixture on a machine tool	20	-	-	-
NOS Total	100	-	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6905
NOS Name	Design and Fabricate Jigs and Fixtures
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, CNC Programmer
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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TNSD/CSC/N6906: Execute Quality Control and CMM Inspections

Description

Able to Understand the modern quality management systems Able to do basic measuring on CMM

Scope

The scope covers the following :

- Understand SQC and Quality Management Systems.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn Statistical quality control
- PC2.** Understand Quality Management Systems
- PC3.** Understand Total Quality Management tools
- PC4.** Learn the Co-ordinate Measuring Machine (CMM)

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn Statistical quality control
- KU2.** Learn the Co-ordinate Measuring Machine (CMM)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Understand Quality Management Systems

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Learn Statistical quality control	25	-	-	-
PC2. Understand Quality Management Systems	25	-	-	-
PC3. Understand Total Quality Management tools	25	-	-	-
PC4. Learn the Co-ordinate Measuring Machine (CMM)	25	-	-	-
NOS Total	100	-	-	-

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6906
NOS Name	Execute Quality Control and CMM Inspections
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, CNC Programmer
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N6907: Apply Principles of Strength of Materials in Design

Description

Understand the effect of stress, strain Understand the effect of temperature on parts/structures.

Scope

The scope covers the following :

- Understand the concept of stress and strain.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the Relationship of Stress and strain
- PC2.** Understand the importance of thermal effect on Stress and Strain
- PC3.** Learn to Calculate the stress and strain based on load conditions
- PC4.** Understand the Effect of load on struts and columns
- PC5.** Learn to Calculate the deflection due to bending
- PC6.** Understand the effect of torsion on shafts
- PC7.** Understand the effect of torsion on springs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the importance of thermal effect on Stress and Strain.
- KU2.** Understand the Effect of load on struts and columns

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn the Relationship of Stress and strain.
- GS2.** Learn to Calculate the stress and strain based on load conditions

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Learn the Relationship of Stress and strain	10	-	-	-
PC2. Understand the importance of thermal effect on Stress and Strain	10	-	-	-
PC3. Learn to Calculate the stress and strain based on load conditions	30	-	-	-
PC4. Understand the Effect of load on struts and columns	10	-	-	-
PC5. Learn to Calculate the deflection due to bending	20	-	-	-
PC6. Understand the effect of torsion on shafts	10	-	-	-
PC7. Understand the effect of torsion on springs	10	-	-	-
NOS Total	100	-	-	-

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Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6907
NOS Name	Apply Principles of Strength of Materials in Design
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N6908: Estimate and Cost Manufacturing Projects

Description

Able to find out the approximate/actual cost of products by costing procedures. Able to calculate the cost of components done by different machining operations

Scope

The scope covers the following :

- Understand the difference between estimating and costing
- Understand the different components which constitute the cost

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Learn the importance and difference between estimation and costing

PC2. Familiar with Components of the cost and selling price of an item

PC3. Understand the Basic machine hour rate

PC4. Understand the Terminology used in business transactions

PC5. Learn to Calculate cost of machined components

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Basic machine hour rate

KU2. Understand the Terminology used in business transactions

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Familiar with Components of the cost and selling price of an item

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Learn the importance and difference between estimation and costing	15	-	-	-
PC2. Familiar with Components of the cost and selling price of an item	20	-	-	-
PC3. Understand the Basic machine hour rate	25	-	-	-
PC4. Understand the Terminology used in business transactions	20	-	-	-
PC5. Learn to Calculate cost of machined components	20	-	-	-
NOS Total	100	-	-	-

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Not
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National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6908
NOS Name	Estimate and Cost Manufacturing Projects
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

TNSD/CSC/N6909: Implement Production Planning and Control Strategies

Description

Familiarize with the systems followed in machine shop/tool room.

Scope

The scope covers the following :

- Understand the functioning of Material management.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Concepts of Material management
- PC2.** Learn Evaluation and Documentation
- PC3.** Understand the Ergonomics and Accident prevention
- PC4.** Learn to Evaluate machining time and costing
- PC5.** Understand the Job sequence for processes and Time study
- PC6.** learn the PPAP documents

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the Concepts of Material management
- KU2.** Understand the Ergonomics and Accident prevention

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn to Evaluate machining time and costing
- GS2.** learn the PPAP documents

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Understand the Concepts of Material management	15	-	-	-
PC2. Learn Evaluation and Documentation	15	-	-	-
PC3. Understand the Ergonomics and Accident prevention	10	-	-	-
PC4. Learn to Evaluate machining time and costing	20	-	-	-
PC5. Understand the Job sequence for processes and Time study	30	-	-	-
PC6. learn the PPAP documents	10	-	-	-
NOS Total	100	-	-	-

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6909
NOS Name	Implement Production Planning and Control Strategies
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, CNC Programmer
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N6910: Apply Principles of Management in a Manufacturing Environment

Description

Familiarize with the different functions of management which are required for a successful organization

Scope

The scope covers the following :

- Understand the functioning of different departments in an organization.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Importance of Management and Organization

PC2. Understand the importance of planning, coordinating & Controlling

PC3. Understand the Importance of staffing

PC4. Understand to differentiate between Company and Trade

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Importance of Management and Organization

KU2. Understand the importance of planning, coordinating & Controlling

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Understand to differentiate between Company and Trade

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Understand the Importance of Management and Organization	25	-	-	-
PC2. Understand the importance of planning, coordinating & Controlling	25	-	-	-
PC3. Understand the Importance of staffing	25	-	-	-
PC4. Understand to differentiate between Company and Trade	25	-	-	-
NOS Total	100	-	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6910
NOS Name	Apply Principles of Management in a Manufacturing Environment
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, CNC Programmer
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0501: OJT(INDUSTRY)

Description

Understand the work culture and procedure followed in a CNC Machine Shop/Industry for executing CNC Programmer Job on Multi-Axial Machines/Projects.

Scope

The scope covers the following :

- Understand the work culture and procedure followed in a CNC Machine Shop/Industry for executing CNC Programmer Job on Multi-Axial Machines/Projects.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Perform a project using various programs in automation industry to make the system free from errors and self-sustainable.	-	80	-	20
NOS Total	-	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0501
NOS Name	OJT(INDUSTRY)
Sector	Capital Goods
Sub-Sector	
Occupation	CNC Programmer, CNC Programmer
NSQF Level	4.5
Credits	10
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping TNSDC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- TNSDC monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

Qualification Pack

- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

On the Job:

1. Each module will be assessed separately.

2. The candidate must score 60% in each module to successfully complete the OJT.

3. Tools of Assessment that will be used for assessing whether the candidate is having desired skills and etiquette of dealing with customers, understanding needs & requirements, assessing the customer and perform Soft Skills effectively:

- Videos of Trainees during OJT

4. Assessment of each Module will ensure that the candidate is able to:

- Effective engagement with the customers
- Understand the working of various tools and equipment

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TNSD/CSC/N6901.Perform Advanced Workshop Operations	-	100	75	25	200	15
TNSD/CSC/N6902.Implement Advanced CNC Programming Techniques	100	90	-	10	200	10
TNSD/CSC/N6903.Generate Complex CNC Toolpaths using Master cam	-	85	-	15	100	10
TNSD/CSC/N6904.Design Advanced 3D Models using SolidWorks	-	100	-	-	100	10
TNSD/CSC/N6905.Design and Fabricate Jigs and Fixtures	100	-	-	-	100	10
TNSD/CSC/N6906.Execute Quality Control and CMM Inspections	100	-	-	-	100	10
TNSD/CSC/N6907.Apply Principles of Strength of Materials in Design	100	-	-	-	100	5
TNSD/CSC/N6908.Estimate and Cost Manufacturing Projects	100	-	-	-	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
TNSD/CSC/N6909.Implement Production Planning and Control Strategies	100	-	-	-	100	5
TNSD/CSC/N6910.Apply Principles of Management in a Manufacturing Environment	100	-	-	-	100	5
TNSD/CSC/N0501.OJT(INDUSTRY)	-	80	-	20	100	10
Total	720	485	75	70	1350	100

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Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.